

Cover Letter - Geomatics, Natural Hazards and Risk

To the Senior Editor,

Geomatics, Natural Hazards and Risk

Taylor & Francis

Dear Editor,

We submit our original research manuscript titled 'RiskPulse: An Integrated Research Software Architecture for Multi-Hazard Evidence Governance, Remote Sensing, and Decision Support in Mountainous Terrain' for consideration for publication in Geomatics, Natural Hazards and Risk.

Mountain hazard management in high-relief catchments such as the Indian Himalayas requires synthesizing disparate hydrometeorological observations, multispectral remote sensing, hydrodynamic solver outputs, and OSINT incident reports. RiskPulse addresses this architectural challenge by coupling multi-source evidence acquisition with deterministic calculation engines, explicit provenance tracking, and a Controlled Promotion Gate that strictly isolates research outputs from operational decision tools.

Within our audited software scope (348/348 passing unit/integration tests), all computational pathways and software contracts are fully verified. We transparently document the uncalibrated global reference status of the Caine landslide threshold ($I = 14.82 * D^{-0.39}$) and Rational flood model ($C=0.65$), explicitly distinguishing software verification from empirical field validation.

The author confirms that this manuscript is original, has not been published previously in whole or in substantial part, and is submitted exclusively to Geomatics, Natural Hazards and Risk. The author declares no conflicts of interest, and this research received no external funding.

Sincerely,

Kuldeep Singh

Faculty, Department of Interdisciplinary Studies

**Himachal Pradesh University, Shimla, Himachal Pradesh,
India**

Email: kuldeepsinghihs@gmail.com | Phone: +91 7018431329